Name: _____ Period: _____

IN CLASS LAB 3 ALTERNATE

Rock Cycle

How do Earth's internal and surface processes transform rocks from one type to another over time?

LEARNING INTENTION

We are learning how Earth's surface and internal processes transform rocks from one type to another so that we can use a model to explain how matter cycles through the rock cycle.

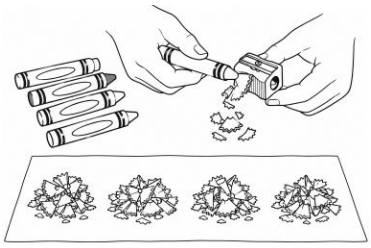
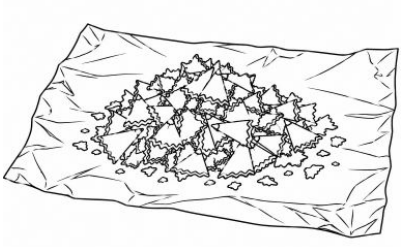
SUCCESS CRITERIA

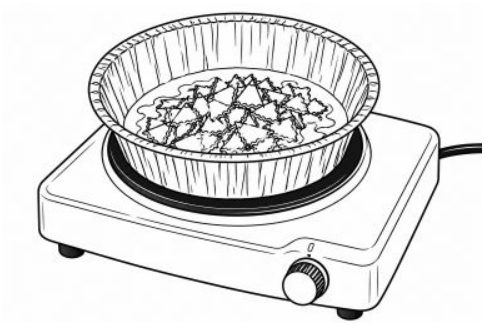
- I can model how rocks change through weathering, erosion/deposition, compaction, heat and pressure, melting, and cooling.
- I can explain how each step of the crayon model represents a real rock cycle process.
- I can identify strengths and limitations of the model compared with the real rock cycle.

These also line up directly with the post-lab questions, especially where students match the modeled

Part 1: Lab Summary

In class, your classmates modeled the rock cycle using crayons as their "rock material." Read what they did and what they observed at each step, and study each image. Each step represents a real Earth process. Use this summary to answer the Post Lab questions.

Step 1: Mechanical Weathering **What students did:** Shaved crayons with a pencil sharpener and made a separate small pile of shavings for each color. **What students observed:** Small, loose pieces of wax. Each color stayed in its own pile.	
Step 2: Erosion and Deposition **What students did:** Moved one color of shavings onto a piece of aluminum foil, then stacked the other colors on top one at a time. Folded the foil into a packet about 4 x 4 inches. **What students observed:** Separate layers of color stacked on top of each other. The pieces were still loose.	
Step 3: Compaction and Cementation (Lithification) **What students did:** Pressed on the foil packet by hand, then opened it and tried to pick up a piece. **What students observed:** The shavings stuck together loosely and the colored layers were still visible. The "rock" was weak and crumbled easily when picked up.	

Step 4: Heat and Pressure (Metamorphism) **What students did:** Rewrapped the pieces in foil, stood on the packet, and pressed down firmly with their body weight. Then they broke the "rock" in half to look inside. **What students observed:** The pieces were pressed into a more solid block that was harder to break. The layers were still there but were squeezed thinner and bent.	
Step 5: Melting, Cooling, and Solidification (Igneous Formation) **What students did:** Placed the pieces in an aluminum bowl. The teacher heated the bowl until the wax melted into "magma." The melted material was then cooled in an ice-water bath. **What students observed:** The colors flowed and blended together. After cooling, the wax was one solid piece with no separate layers.	

Post Lab:

1. Match each lab step to the correct Earth process modeled.

Write the letter of the correct Earth process (A–E) on the line next to each numbered lab step. Each answer will be used only once.

_____1. Shaving crayons

_____2. Layering shavings

_____3. Pressing layers by hand

_____4. Standing on the packet

_____5. Melting & cooling

A. Heat & pressure
B. Compaction & cementation
C. Mechanical weathering
D. Melting & solidification
E. Erosion & deposition

2. A crayon shaving was layered, pressed, pressed harder with body weight, and then melted.

a. Identify the sequence of rock types this process modeled.

Sediment → _____ → _____ → _____

b. Explain why each change occurred. Use at least one observation from Part 1 as evidence.

_____

_____

_____

3. The crayon activity was used as a model of the rock cycle.

a. Describe two ways the crayon model accurately represents real Earth processes.

1. _____

2. _____

b. Describe two limitations of the crayon model compared with real rock cycle processes.

1. _____

2. _____

4. In the real world, rock cycle processes involve interactions between Earth's spheres. Choose one rock cycle process from the model and explain how it involves at least **two** Earth spheres (geosphere, hydrosphere, atmosphere, biosphere).

_____

_____

_____

5. In the crayon model, each change happened in a few minutes. Explain why the same changes take much longer in the real rock cycle.

_____

_____

_____

6. Granite forms when magma cools slowly beneath Earth's surface. Millions of years later, the granite is exposed at the surface, breaks into sediments, and eventually forms sandstone. Explain the sequence of rock cycle processes that caused this change.

_____

_____

_____

_____

Cluster Set:

Base your answer(s) to the following question(s) 1-7 on the Earth Science Reference Tables, the diagram below, and your knowledge of Earth science. The photograph shows examples of older metamorphic rocks and younger volcanic rocks found in Yosemite National Park. The image on the left shows an outcrop of glacially polished quartzite interlayered with schist at May Lake. The image on the right shows Little Devils Postpile, an 8-million-year-old columnar-jointed andesite formation located near Tuolumne Meadows. Some questions may require the use of the *2024 Edition Reference Tables for Earth and Space Sciences.*

Yosemite National Park: A Landscape Shaped Over Time

Visitors to Yosemite National Park see towering rock walls, deep valleys, winding rivers, and powerful waterfalls. The cliffs rise sharply from the valley floor, and many surfaces appear smooth and curved, while other areas contain layered rock or broken fragments along the base of slopes.

Scientists have determined that the rocks exposed in Yosemite formed deep underground long before they were visible at the surface. Over time, forces within Earth pushed these rocks upward. Once exposed, rain, snow, rivers, and ice began to reshape the landscape. Water moved through cracks in the rock, sometimes freezing and expanding. Large masses of ice once covered parts of the region, carving valleys and transporting rock material to new locations.

In some lower areas, loose sediments accumulated and were later buried beneath additional material. In other places, rocks that were carried deeper underground experienced increased heat and pressure. Today, different types of rock can be found throughout the park, each recording part of Yosemite's geologic history.

The landscape of Yosemite continues to change as Earth's internal processes and surface processes interact.



_____1. Rocks in Yosemite were described as forming deep underground before becoming visible at the surface. Which type of rock most likely formed this way?

(1) Extrusive igneous rock
(2) Intrusive igneous rock
(3) Sedimentary rock
(4) Metamorphic rock

_____2. Which process most directly caused rock material in Yosemite to melt beneath Earth's surface?

(1) Energy from the Sun
(2) Surface water movement
(3) Heat from Earth's interior
(4) Plant root growth

_____3 Which statement from the passage best supports this claim?

Claim: Forces within Earth played a major role in shaping Yosemite's landscape.

(1) Visitors see waterfalls flowing through the park.
(2) Rocks were formed deep underground and later pushed upward by forces within Earth.
(3) Large cliffs rise from the valley floor.
(4) Sediments accumulated in lower areas.

4. The passage describes rocks forming deep underground and later being pushed upward over millions of years. Explain how tectonic plate movement relates to the **temporal** scale of this process.

5. The passage describes rocks forming deep underground and later becoming exposed at Earth's surface.
Identify the process responsible for this vertical movement and explain how it represents a change in **spatial** scale within Earth's system.

6. Describe one interaction between the geosphere and another Earth sphere that is described in the passage.

7.Based on the passage and your knowledge of Earth science, complete the pathway below by selecting the correct row from the options below.

> A rock in Yosemite originally formed as an intrusive igneous rock when _____ (A) cooled slowly beneath Earth's surface. Over time, tectonic plate movement caused _____ (B), which changed the rock's depth and brought it closer to the surface. If the rock were later buried deeper, increased heat and pressure could change it into _____ (C).

	(A)	(B)	(C)
(1)	Magma	Deposition	Sedimentary Rock
(2)	Magma	Uplift	Metamorphic Rock
(3)	Lava	Deposition	Metamorphic Rock
(4)	Lava	Uplift	Sedimentary Rock